

EVIDENCE-BACKED EXPORT COMPLIANCE

From semiconductor datasheet to human-review- ready ECCN memo.

Substrata turns scattered technical evidence into cited review paths, explicit uncertainty, and an audit-ready classification memo draft.

- Source-grounded facts
- Recommended review paths
- Human approval

A HIGH-STAKES WORKFLOW HELD TOGETHER BY DOCUMENTS

The bottleneck is not finding an answer. It is building a **defensible review record**.

01 Evidence is fragmented

Datasheets, old memos, spreadsheets, email, and reviewer notes sit in separate systems.

02 Technical work gets repeated

Reviewers reconstruct device facts and prior reasoning for every new product or variant.

03 Uncertainty is hard to audit

Missing facts, reviewer questions, and the basis for a recommendation are easily lost.

WHAT TEAMS MUST PRODUCE

A **review-ready** evidence package

Not a black-box label: a source-grounded workup a qualified reviewer can inspect, challenge, and approve.

Classification memo draft

NEEDS HUMAN REVIEW

Extracted facts

Citations

Review paths

Uncertainty flags

Reviewer questions

Audit trail

A REVIEW WORKUP—NOT A BLACK-BOX ANSWER

Upload a datasheet. Receive a cited classification memo draft.

Extracted specifications

Product form, architecture, interfaces, memory, security, and performance.

Candidate ECCNs

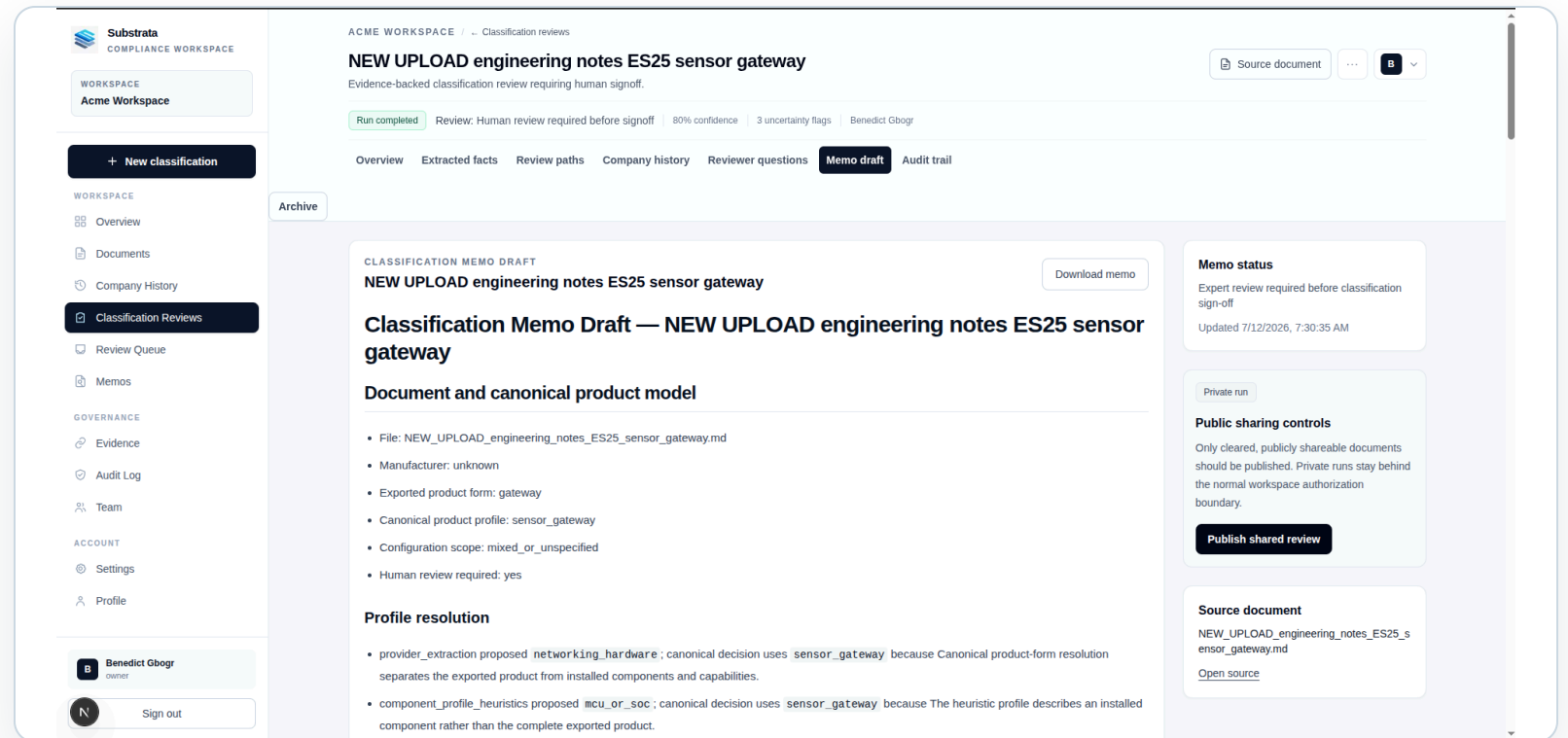
Recommended review paths with why they may or may not apply.

Citations + uncertainty

Source evidence, missing facts, conflicts, and reviewer questions.

Human-review-ready memo

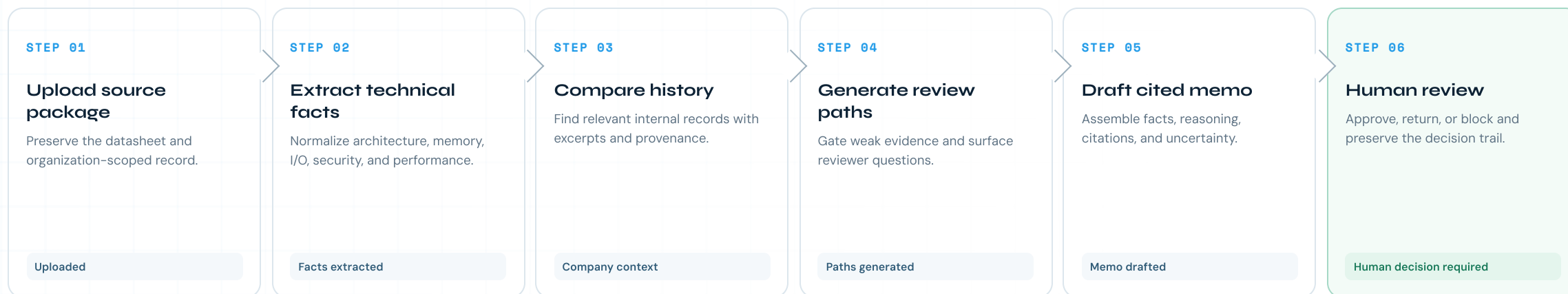
A draft tied to evidence and its audit trail; human signoff remains mandatory.



The screenshot displays the Substrata Compliance Workspace interface. On the left is a sidebar with navigation options: Overview, Documents, Company History, Classification Reviews (selected), Review Queue, Memos, Evidence, Audit Log, Team, Settings, and Profile. The main content area is titled 'ACME WORKSPACE / Classification reviews' and shows a 'NEW UPLOAD engineering notes ES25 sensor gateway' review. A status bar indicates 'Run completed', 'Review: Human review required before signoff', '80% confidence', '3 uncertainty flags', and 'Benedict Gbogor'. Below this are tabs for Overview, Extracted facts, Review paths, Company history, Reviewer questions, Memo draft (selected), and Audit trail. The 'Memo draft' tab shows a 'CLASSIFICATION MEMO DRAFT' for the same title, with a 'Download memo' button. The memo content includes a 'Document and canonical product model' section with a list of items (File, Manufacturer, Exported product form, Canonical product profile, Configuration scope, Human review required) and a 'Profile resolution' section with two bullet points explaining the resolution process. On the right side of the interface, there are three panels: 'Memo status' (Expert review required before classification sign-off, Updated 7/12/2026, 7:30:35 AM), 'Public sharing controls' (Only cleared, publicly shareable documents should be published. Private runs stay behind the normal workspace authorization boundary. with a 'Publish shared review' button), and 'Source document' (NEW_UPLOAD_engineering_notes_ES25_sensor_gateway.md with an 'Open source' button).

DATASHEET IN. REVIEW-READY WORKUP OUT.

A structured compliance workflow—not a chatbot.



For semiconductor and advanced-hardware compliance teams

Category 3 · Category 5 Part 2 · adjacent review paths

DISCOVERY IS SHAPING A FUNCTIONING PRODUCT

Early Validation and Product Progress

CUSTOMER DISCOVERY FUNNEL

~200

Targeted outreach emails sent

Trade compliance, export controls, semiconductors, advanced hardware, and related fields.

~10

Active industry discussions

Relevant professionals helping validate the problem and refine the product.

Next

Potential design partners being identified

Discussions may lead to design-partner or adviser relationships; none are presented as formal commitments.

Practitioner input already changed the product. Feedback from Anna Goncz directly influenced the company-history and precedent-based classification workflow. She is a potential design partner or industry adviser in discussion, not a confirmed partner.

Functioning product workflow

Technical-spec extraction, ECCN candidates, supporting citations, uncertainty flags, company-history comparison, audit trail, human review, and memo generation.

Technical extraction

ECCN candidates

Citations + flags

Company history

Human review

Memo + audit trail

72 / 72

Lifecycle integration

18 / 18

API regression

32 / 32

Worker tests

What the discussions are doing

Validating the problem, refining the product, and identifying potential design-partner relationships.

A REVIEWER-SAFE INTELLIGENCE ARCHITECTURE

AI captures the evidence. Deterministic logic controls the path. Humans own the decision.

AI

Model-assisted extraction

Turns long technical documents into normalized, source-linked facts.

- Architecture and interfaces
- Memory and performance
- Cryptography and security
- Source snippets and confidence



f

Deterministic review engine

Controls candidate routing, evidence gates, contradictions, and fallback behavior.

- Rules-version provenance
- Missing-evidence checks
- Blocked vs. active candidates
- Reproducible review behavior



✓

Human review workspace

Makes reasoning inspectable and turns uncertainty into explicit reviewer work.

- Recommended review paths
- Reviewer questions
- Memo versioning
- Conclusion + audit trail

Why this is different: the model is not the classification authority. The product is built around traceable work, controlled uncertainty, and accountable human signoff.

ACCELERATED DOCUMENT INTELLIGENCE

AMD compute scales the expensive work without weakening reviewer control.

01

Gemma-assisted extraction

Model inference captures technical facts from dense hardware documents and returns structured output.

02

ROCm-capable worker path

A containerized Python worker and stable job contract support AMD GPU execution.

03

Batch-ready architecture

Worker isolation prepares company-history ingestion and document batches for accelerated execution.

INPUT

Source package

Datasheet text + run context



AMD EXECUTION

Gemma + ROCm path

Extraction and normalization



ARTIFACT

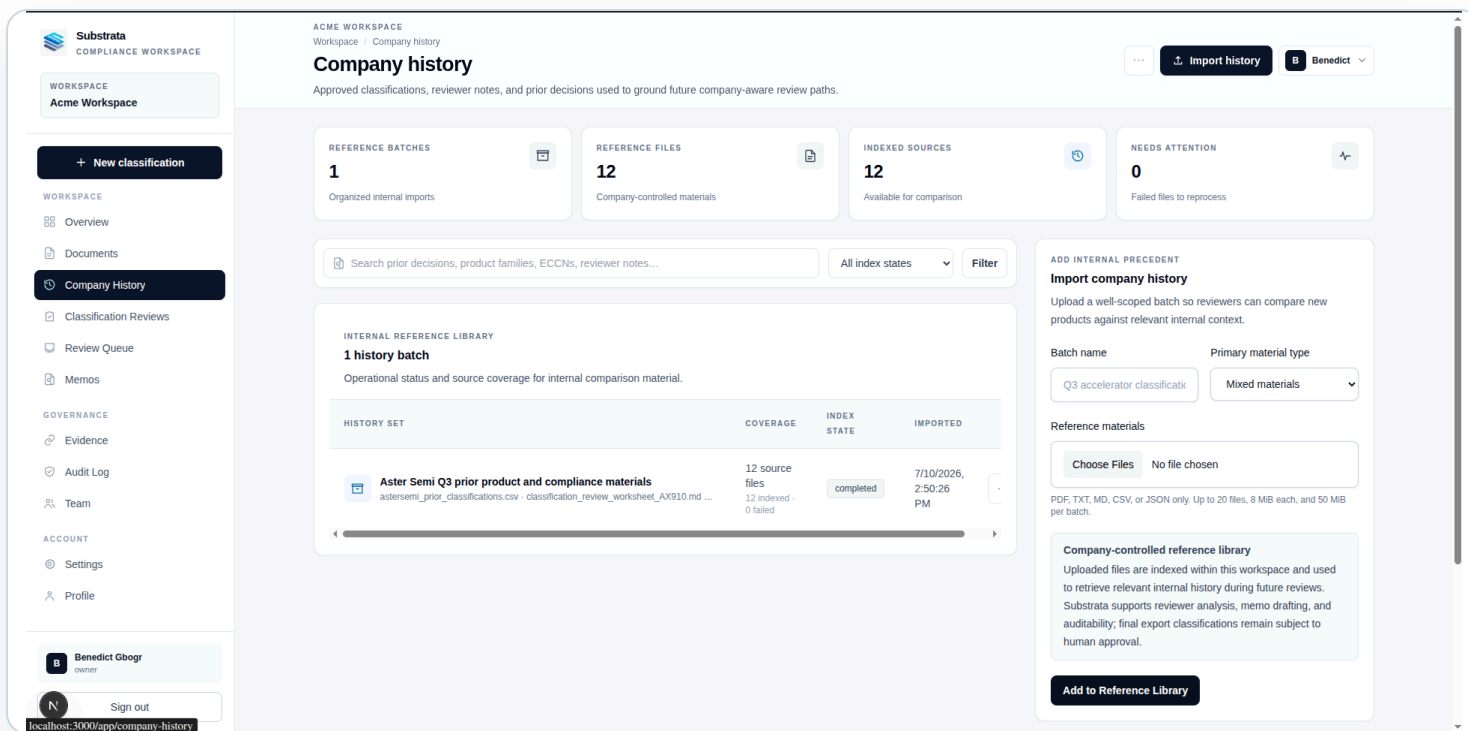
Structured evidence

Facts, snippets, confidence, provenance

Clear authority boundary: accelerated inference proposes structured facts; the deterministic engine gates review paths; a qualified human approves the workup.

A FOCUSED WEDGE INTO A DURABLE SYSTEM OF RECORD

Start with **datasheet-to-ECCN** review. Expand with every approved workup.



The screenshot displays the Substrata Compliance Workspace interface. The left sidebar shows navigation options: Workspace (Acme Workspace), New classification, Overview, Documents, Company History (selected), Classification Reviews, Review Queue, Memos, Governance (Evidence, Audit Log, Team), and Account (Settings, Profile). The main content area is titled 'ACME WORKSPACE' and 'Company history'. It features a top bar with 'Import history' and 'Benedict' dropdown. Below this are four summary cards: 'REFERENCE BATCHES' (1), 'REFERENCE FILES' (12), 'INDEXED SOURCES' (12), and 'NEEDS ATTENTION' (0). A search bar and filter options are present. The 'INTERNAL REFERENCE LIBRARY' section shows a table with columns: HISTORY SET, COVERAGE, INDEX STATE, and IMPORTED. The table lists a batch named 'Aster Semi Q3 prior product and compliance materials' with 12 source files, 12 indexed, 0 failed, and a completion status of 'completed' on 7/10/2026 at 2:50:26 PM. The right sidebar contains sections for 'ADD INTERNAL PRECEDENT' (Import company history), 'Reference materials' (Choose Files), and 'Company-controlled reference library' (Add to Reference Library).

Beachhead buyer

In-house trade compliance managers and analysts at semiconductor and advanced-hardware companies.

Recurring trigger

New products, variants, supplier documents, acquisitions, and customer requests.

Expansion engine

Each approved workup strengthens a reusable, organization-scoped classification-history layer.

Commercial model

B2B SaaS priced by organization, review volume, users, and enterprise deployment requirements.

THE WORKFLOW HAS CROSSED AN INFLECTION POINT

Hardware complexity is rising faster than compliance teams can add reviewers.

01

Technical products are harder to review

Advanced compute, mixed-function devices, security features, and variants create more facts and more possible review paths.

02

Institutional knowledge is trapped

Prior decisions exist, but they are difficult to find, compare, and reuse with the provenance reviewers need.

03

AI can structure the work—now

Modern open models can extract technical evidence at scale when paired with deterministic controls and human approval.

The opportunity

Make the review-ready workup—not a black-box answer—the durable operating layer for hardware export compliance.

WHAT THE PRE-SEED ROUND UNLOCKS

\$750K provides 15–18 months to reach paid enterprise pilots.

NOW • WORKING MVP

End-to-end review workup

- Source upload and extraction
- Cited review paths
- Company-history comparison
- Memo, human review, audit trail
- Tested lifecycle and security controls

15–18 MONTHS • BUILD

Production and validation

- Production-grade product and security
- Expanded classification and rule coverage
- Expert-reviewed validation
- Managed AMD execution path
- Enterprise workflow controls

ROUND TARGETS • PROVE

Design partners and pilots

- 5–10 hardware design partners
- 3–5 paid enterprise pilots
- Measured review-time impact
- Consistency and traceability evidence
- Audit-readiness evidence

Raising a \$750,000 pre-seed round

Complete production readiness, work with trade-compliance experts, onboard semiconductor and advanced-hardware design partners, and convert early validation into paid pilots.

PRODUCT INSIGHT MEETS EXECUTION INFRASTRUCTURE

Two builders turning a high-stakes workflow into dependable software.



Benedict Gbogr

CO-FOUNDER · AI INFRASTRUCTURE & ENGINEERING

Founder of Jungle Grid, an execution layer for AI agents and GPU workloads. Benedict builds the orchestration, container execution, artifacts, and reliability infrastructure behind Substrata's document intelligence.

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Nathan Ulanja

CO-FOUNDER · PRODUCT & COMPLIANCE WORKFLOW

Nathan originated Substrata's initial product direction: helping semiconductor and advanced-hardware teams turn manual export-control research into cited, human-review-ready workups.

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Built together before. Building Substrata together now.

Product workflow, full-stack execution, and AI infrastructure combined in one focused founding team.

THE COMPLIANCE DECISION LAYER FOR ADVANCED HARDWARE

Evidence-backed workups. Human-approved decisions.

Substrata combines accelerated document intelligence with traceable evidence, explicit uncertainty, company context, and accountable human review.

Raising \$750K to turn Substrata into the classification system of record for advanced-hardware companies.

15–18 months runway

Target: 5–10 design partners

Target: 3–5 paid pilots

Production readiness